WEEK 4 MACHINE LEARNING PROJECT

Multiple Linear Regression - House Price Prediction

Overview

You have been given a housing dataset containing approximately 2,000 property records. Unlike a perfectly prepared classroom dataset, this file contains data-quality problems that must be investigated and corrected before it is suitable for Machine Learning.

Your task is to preprocess the data, build a Multiple Linear Regression model that predicts house prices from several property features, evaluate the model, visualize its performance, make new predictions, and organize the completed project professionally on GitHub.

Project Scenario

A property company wants a simple model that can estimate the price of a house from information such as its size, location, number of bedrooms and bathrooms, garage capacity, age, distance from the city centre, property type, and garden availability. The company has supplied historical records, but the dataset is not completely clean.

Dataset

File: Week_4_Messy_Housing_Data_2000.csv

Column	Meaning
Property_ID	Identifier for each property record.
Location	City/location of the property.
Property_Type	Apartment, bungalow, duplex, detached house, terrace, etc.
Size_SqFt	Approximate property size in square feet.
Bedrooms	Number of bedrooms.
Bathrooms	Number of bathrooms.
Garage_Spaces	Number of available garage spaces.
Property_Age_Years	Approximate age of the property.
Distance_City_Center_Km	Distance from the city centre in kilometres.
Has_Garden	Whether the property has a garden.
House_Price	Target variable: the house price to be predicted.

Important: The dataset intentionally contains missing values, duplicate records, inconsistent category labels, incorrect data types, text inside numerical columns, and a small number of suspicious or implausible values. You are expected to discover and handle these issues yourself.

PHASE 1 - DATA INSPECTION & PREPROCESSING

Before training any model, investigate the dataset carefully. Do not immediately delete every problematic row. Decide what each problem means and choose a reasonable preprocessing strategy.

Requirements

1. Load the CSV file with Pandas and inspect its shape, columns, data types, sample rows, descriptive statistics, missing values, and duplicate records.
2. Identify the data-quality problems in the dataset. Document the important problems you find using Markdown cells in your notebook.
3. Handle duplicate records appropriately.
4. Handle missing values using reasonable decisions. Different columns may require different approaches.
5. Clean numerical columns that contain text or incorrect data types and convert them to suitable numerical types.
6. Standardize inconsistent categorical values such as location, property type, and garden information.
7. Investigate suspicious or implausible numerical values and decide how they should be treated.
8. Check the cleaned dataset again before beginning Machine Learning.

PHASE 2 - PREPARE THE DATA FOR MACHINE LEARNING

9. Define the feature matrix X and the target y. The target is House_Price.
10. Remove Property_ID from the model features unless you can justify a meaningful reason for using it.
11. Convert categorical features into a numerical form that a Linear Regression model can use. Research an appropriate encoding technique.
12. Split the prepared data into training and testing sets using train_test_split.
13. Clearly show the shape of your training and testing data.

Research point: A Machine Learning model cannot directly interpret labels such as "Awka", "Duplex", or "Yes" as ordinary numerical features. Find out how categorical variables can be encoded appropriately for regression.

PHASE 3 - BUILD THE MULTIPLE LINEAR REGRESSION MODEL

14. Create a LinearRegression model using Scikit-learn.
15. Train the model using the training data with .fit().
16. Use the trained model to predict house prices for the test data.
17. Display a small comparison of actual house prices and predicted house prices.

This is Multiple Linear Regression because the model uses several input features to predict one continuous target: House_Price.

PHASE 4 - MODEL EVALUATION

Evaluate the model using all three metrics below:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R-squared (R^2)

Do not only print the numbers. Write a short explanation of what each metric tells you about the model and interpret your own results.

Important Note About the Metrics

MSE and RMSE are error measures, so lower values generally indicate smaller prediction errors. R^2 measures how much of the variation in the target is explained by the model, with higher values generally indicating a better fit. Because these metrics use different scales, a single bar chart of their raw values may be misleading.

You may create a compact evaluation figure that displays MSE, RMSE, and R^2 together as labelled metric cards, annotations, or separate panels. If you choose a bar chart, clearly explain the scale difference rather than treating the bar heights as directly comparable.

PHASE 5 - MODEL PERFORMANCE VISUALIZATIONS

Create charts that help you understand how well the model learned from the data. Your project should contain at least the following visualizations:

18. Actual vs Predicted Prices: Create a scatter plot comparing actual test prices with the prices predicted by the model. Include a reference line representing perfect predictions if you know how to do so.
19. Residual/Error Plot: Visualize the prediction errors (actual price minus predicted price) so you can inspect whether the model tends to overpredict or underpredict particular houses.
20. Model Evaluation Summary: Present MSE, RMSE, and R^2 visually in a clear evaluation figure. Because their scales differ, choose a presentation that does not falsely imply that the raw values are directly comparable.

For every major chart, include a clear title, axis labels where applicable, and a short Markdown interpretation beneath the chart.

PHASE 6 - PREDICT NEW HOUSE PRICES

After the model has been trained and evaluated, use it to estimate prices for new houses that were not part of the dataset.

21. Create at least three imaginary houses with different combinations of size, location, bedrooms, bathrooms, garage spaces, property age, distance, property type, and garden availability.
22. Apply the same preprocessing/encoding approach used during training before sending new data to the model.
23. Use the trained model to predict the price of each house.
24. Display the house details together with the predicted price in a readable format.

PHASE 7 - SHORT MODEL INTERPRETATION

Write a concise Markdown section answering the following questions:

- Which features were used to predict house price?
- What preprocessing problems did you discover and how did you handle them?
- What do your MSE, RMSE, and R^2 results suggest about model performance?

- From the actual-vs-predicted and residual plots, where does the model appear to perform well or poorly?
- What limitations does this model or dataset have?
- What could you try in the future to improve the model?

PHASE 8 - PROJECT STRUCTURE & GITHUB

Keep the project organized and publish it to GitHub. Use multiple meaningful commits rather than uploading everything only at the end.

Suggested Folder Structure

```
week-4-house-price-regression/
|
|-- data/
|  |-- Week_4_Messy_Housing_Data_2000.csv
|  `-- cleaned_housing_data.csv
|
|-- charts/
|  |-- actual_vs_predicted.png
|  |-- residual_plot.png
|  `-- model_metrics.png
|
|-- house_price_regression.ipynb
|-- README.md
|-- requirements.txt
`-- .gitignore
```

GitHub Requirements

- A clear README.md containing the project objective, dataset description, preprocessing performed, model used, evaluation results, major findings, project structure, and instructions for reviewing/running the notebook.
- A requirements.txt file containing the important Python packages used.
- An appropriate .gitignore file.
- Multiple meaningful Git commits that show the development of the project.
- No passwords, credentials, unnecessary temporary files, or environment folders in the repository.

What Your Project Should Demonstrate

- Data inspection and preprocessing with Pandas
- Handling missing values and duplicates
- Correcting data types and inconsistent categories
- Identifying suspicious values
- Categorical encoding
- Feature/target separation
- Train/test split
- Multiple Linear Regression
- Model training and prediction
- MSE, RMSE, and R^2 evaluation

- Actual-vs-predicted analysis
- Residual/error analysis
- Prediction on new unseen houses
- Clear Markdown explanations
- Professional folder organization
- Git and GitHub workflow

Research Is Allowed

You are expected to research concepts or techniques that you do not already know, including categorical encoding, cleaning mixed-type numerical columns, calculating RMSE, plotting residuals, saving charts, creating requirements.txt, and using .gitignore. You may use documentation, tutorials, search engines, or AI tools, but you must understand and be able to explain every important part of the code you submit.

Submission

- Completed Jupyter Notebook (.ipynb).
- GitHub repository link containing the complete organized project.
- The cleaned dataset may be included in the repository so your preprocessing result can be inspected.

No PDF report is required for this assignment. Your notebook and README should clearly communicate what you did and what you learned.

Important Notes

- Do not train the model before dealing with the major data-quality problems.
- Do not copy a complete solution without understanding it.
- Do not evaluate the model only on the training data.
- Do not assume that a high R^2 automatically means the model is perfect. Inspect prediction errors and limitations too.
- Make sure the notebook can run from beginning to end before submission.

Good luck!

Clean the data carefully, understand your model, evaluate it honestly, and make sure you can explain your decisions.